

Skills Summary

Rational Graphs: Holes, Asymptotes, Intercepts

Use this reference while completing your worksheet or when studying.

1. Cancel first: holes hide in the factors

Hole at $x = c$ when $(x - c)$ cancels top and bottom. Height of the hole: substitute c into the *simplified* function — the original is undefined there, so you cannot substitute into it.

Example: Find the hole of $\frac{x^2 - 16}{x^2 + x - 12}$.

Step 1. The top and bottom are both quadratics, so factor before deciding anything — a shared factor is a hole, a leftover bottom factor is an asymptote.

Step 2. $\frac{(x-4)(x+4)}{(x+4)(x-3)} = \frac{x-4}{x-3}$, so $(x+4)$ cancelled: the hole sits at $x = -4$.

Step 3. Height: $\frac{-4-4}{-4-3} = \frac{-8}{-7} = \frac{8}{7}$.

The hole is at $(-4, \frac{8}{7})$.

Try it: Find the hole of $\frac{x^2 - 25}{x^2 + 2x - 15}$.

($\frac{5}{2}$, $-\frac{1}{2}$) :check

2. Vertical asymptotes: what is left underneath

After cancelling, set each remaining denominator factor equal to zero. Those x -values are vertical asymptotes: near them the graph shoots up or down without bound, and the graph never crosses them.

Example: Vertical asymptotes of $\frac{x-1}{x^2 - 2x - 8}$.

Step 1. Only the denominator can create an asymptote, so factor it and check whether the top shares any of those factors.

Step 2. $x^2 - 2x - 8 = (x-4)(x+2)$, and $x-1$ matches neither, so nothing cancels.

Step 3. Set each factor to zero: $x-4=0$ and $x+2=0$.

$x = 4$ and $x = -2$

Try it: Vertical asymptotes of $\frac{x+2}{x^2 - 9}$.

$x = -2$ and $x = 3$:check

3. Intercepts: where the graph meets the axes

x -intercepts: zeros of the simplified numerator (a cancelled zero is a hole, not an intercept).
 y -intercept: evaluate at $x = 0$ — one substitution, and it exists unless $x = 0$ is itself a hole or an asymptote.

Example: Intercepts of $\frac{x^2 - 4x - 5}{x^2 - 4}$.

Step 1. Two different questions, two different halves of the fraction: the top controls where $y = 0$, and $x = 0$ gives the y -intercept.

Step 2. $x^2 - 4x - 5 = (x - 5)(x + 1)$, which is zero at $x = 5$ and $x = -1$; neither value zeroes $x^2 - 4$, so both are real intercepts.

Step 3. At $x = 0$: $\frac{0 - 0 - 5}{0 - 4} = \frac{-5}{-4} = \frac{5}{4}$.

The y -intercept is $(0, \frac{5}{4})$.

Try it: Intercepts of $\frac{x^2 + x - 6}{x^2 - 1}$.

check: $x = 2, x = -3$ for top; $x = 1, x = -1$ for bottom

4. End behaviour: which degree wins

Compare degrees, top vs bottom: **bottom bigger** $\Rightarrow y = 0$; **equal** $\Rightarrow y = \frac{\text{leading top}}{\text{leading bottom}}$; **top bigger** $\Rightarrow$ no horizontal asymptote.

Example: Horizontal asymptote of $\frac{4x^2 - x}{2x^2 + 5}$.

Step 1. Only the fastest-growing term in each part matters far from the origin, so compare degrees before doing any arithmetic.

Step 2. Both degrees are 2, so take the leading coefficients: $\frac{4}{2} = 2$.

$y = 2$

Try it: Horizontal asymptote of $\frac{3x + 1}{x^2 - 4}$.

check: $x = 0$

(!) **Watch out:** a zero of the denominator is *not* automatically a vertical asymptote — cancel first, or you will call a hole an asymptote. And a graph may cross its horizontal asymptote in the middle; that line only describes the far left and far right.